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PERTRONIC BOWS CONTROLLER

USER MANUAL

AUSTRALIA

Issue 1.2

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1. INTRODUCTION

The Pertronic *BOWS Controller* gets its name from the acronym *BOWS* for Building Occupant Warning System. Used most frequently in Australia, a BOWS system is code-compliant in many smaller building situations when the capability for evacuation by zones is not necessary. (Where evacuation by zones *is* required, a full EWIS (Emergency Warning and Intercommunication System) is appropriate instead of a system using a *BOWS Controller*.)

Pertronic‘s *BOWS Controller* provides a convenient means to manage the Pertronic EVAC Tone Generator, whether the tone generator is used in conjunction with a fire alarm panel or used independently as the building’s evacuation system.

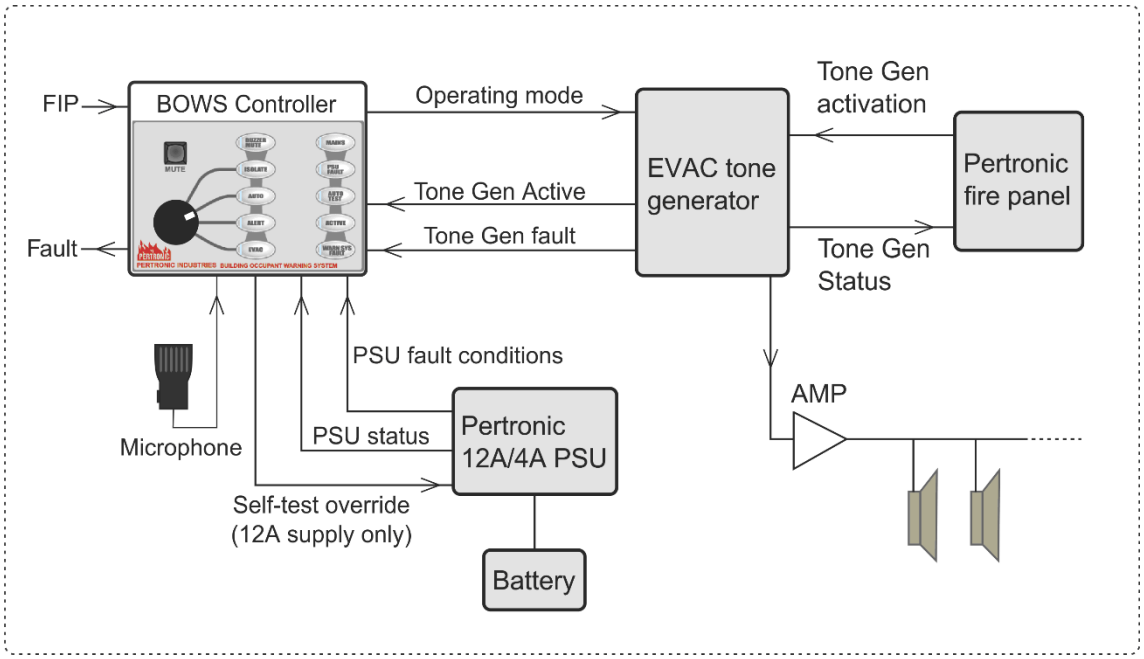
If the EVAC Tone Generator is connected with a Pertronic fire alarm panel, tone generation can be controlled either from the panel’s configuration, or from a *BOWS Controller*. The *BOWS Controller,* along with the associated tone generator and amplifier, can either be mounted in the fire panel or in a standalone cabinet.

The *BOWS Controller* provides an intuitive warning system interface for the operator:

- A clearly-labelled control panel, with
- A four-position, rotary-type Mode Switch,
- LED status and Fault indicators,
- Local fault buzzer, and
- A buzzer *Mute* push-button.

The *BOWS Controller* provides flexible setup options for the installer. Besides monitoring the tone generator, the device also performs the following tasks:

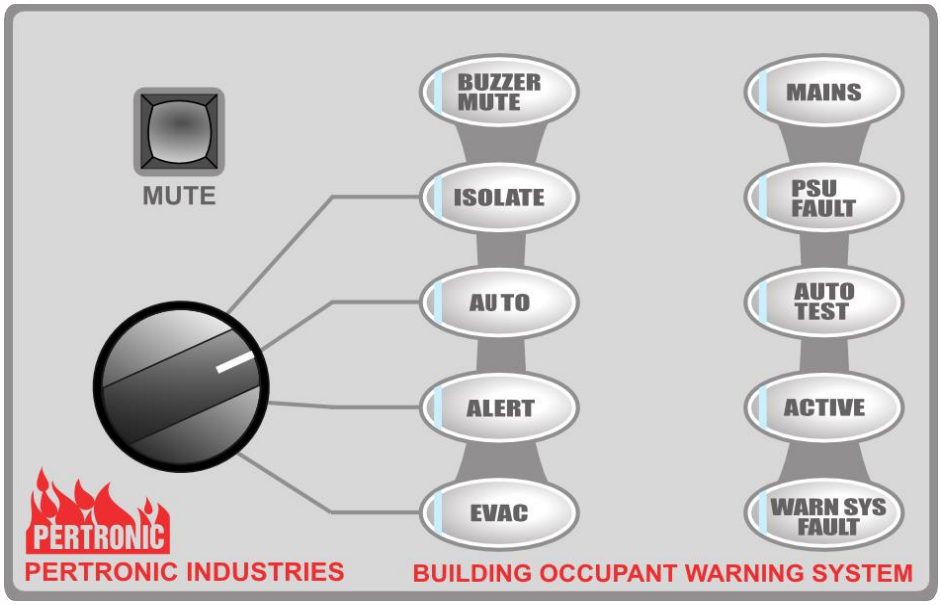
- Monitors power supply and battery status if used with a Pertronic power supply, and
- Permits connecting a variety of appropriate input, output, and control signals, such as those for a P/A system, tone generator, and fire alarm panel.


FIP
BOWS Controller
Operating mode
Tone Gen activation
EVAC tone generator
Pertronic fire panel
Tone Gen Active
Tone Gen fault
Fault
Tone Gen Status
PSU fault conditions
AMP
Microphone
PSU status
Pertronic 12A/4A PSU
Self-test override (12A supply only)
Battery

BOWS System

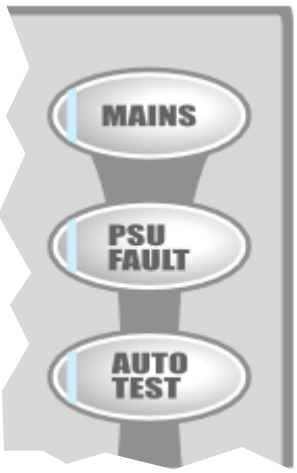
2. OPERATION

The *BOWS Controller* is operated from its front panel, shown below.



BOWS Controller's user panel

2.1 POWER SUPPLY INTERFACE



The *BOWS Controller* is designed to connect directly to a Pertronic 4A or 12A power supply and to indicate the PSU's operating state.

The ***MAINS*** LED shows that the AC supply is active; when operating on battery backup, this LED will be off.

The ***PSU Fault*** LED indicates that a fault condition has been detected by the power supply. Common causes are a missing battery, battery failure, or a charging fault.

The ***AUTO TEST*** LED signifies that the power supply is performing regular "health checks" of the backup battery. During periods of active audio output, the *BOWS Controller* will command the power supply to cease auto-testing of the battery; in this situation the LED will not be lit. (*Note: 12A PSU only*)

2.2 GENERAL OPERATION

The *BOWS Controller* acts as the user interface for a BOWS system. Its operation is described and illustrated in the table beginning on page 6.

2.3 OPERATION—MODE 0

The Mode of operation is set by a MODE switch (SW3) on the BOWS controller PCB.

The BOWS system can be activated via a Pertronic fire panel RS485 connection to the EVAC Tone generator **OR** a direct connection to the FIP connection on the BOWS controller.

When Mode "0" is selected the BOWS system will start with an "Alert" message and after a set delay progress to an "Evac" message.

Activation via EVAC Tone Gen RS485

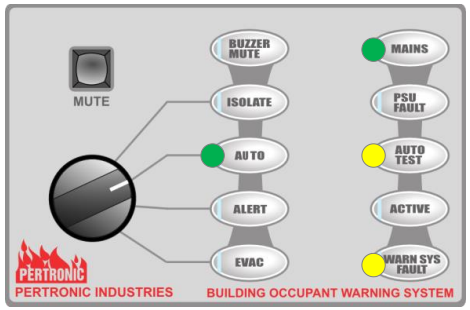
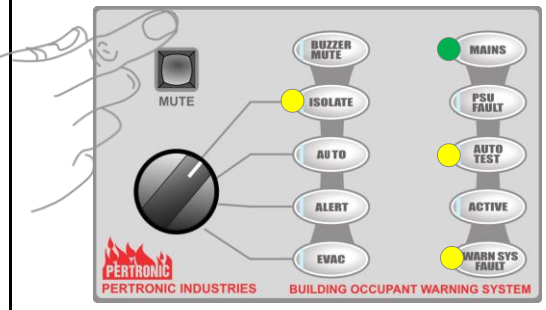
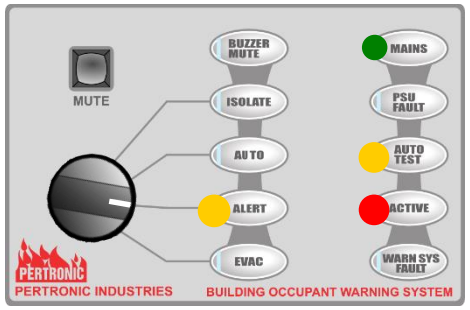
The Pertronic Fire Panel must be programmed to activate the "Alert" message and after a delay the "Evac" message on the EVAC tone generator.

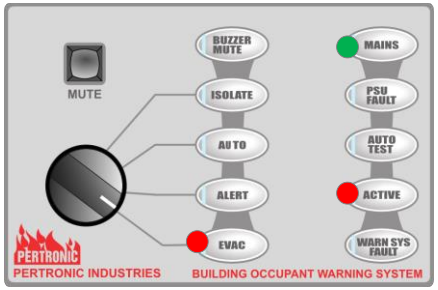
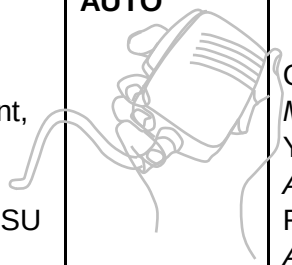
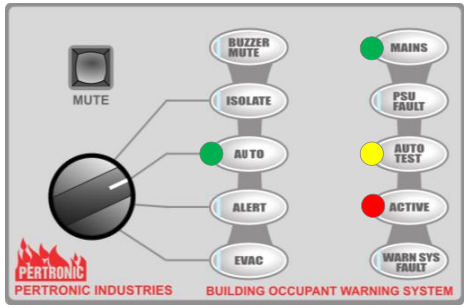
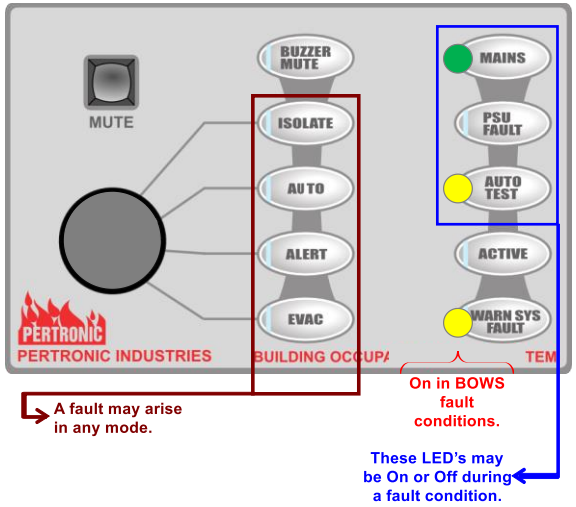
To ensure that the BOWS controller indicates the correct state (Alert or Evac) one of the following delays must be programmed in the Pertronic Fire panel controlling the EVAC Tone generator and the corresponding delay selected using SW4 on the BOWS controller.

DELAY switch position	BOWS Controller EVAC progression delay (SW4)	Fire Panel Config- EVAC delay
0	30 seconds	30 seconds
1	1 minute	1 minute
2	2 minutes	2 minutes
3	3 minutes	3 minutes
4	4 minutes	4 minutes
5	5 minutes	5 minutes
6	6 minutes	6 minutes
7	8 minutes	8 minutes
8	10 minutes	10 minutes
9	Off – stay in Alert	Off – stay in Alert

Activation via BOWS FIP input only

A delay must be set using Delay switch (SW4) as shown in the table above. No Fire Panel programming is required.

Condition:	Description:	Mode Switch:	LEDs display:	Buzzer:	Control Panel LOOKS like this:
Normal Operating mode	Evacuation is armed and ready: Were an alarm event to occur, tones / voice announcements would be automatically initiated per panel's instructions and EVAC tone generator configuration. NOTE: The *BOWS* cabinet door is not to be closed in any mode other than AUTO.	**AUTO**	**Green** for *AUTO* LED. Green for *MAINS* LED & Yellow if *AUTO TEST* is currently active.		
Isolation	Evacuation system is DISABLED: Any alarm triggers or microphone inputs are ignored. The system is NON-OPERATIONAL. Note: If the BOWS cabinet door remains open for more that 8 hours the WARN SYS FAULT LED will flash and the BUZZER will re-sound.	**ISOLATE**	**Yellow** for *ISOLATE* LED. Green for *MAINS* LED, Yellow for *AUTO TEST* Yellow or Yellow-blinking for *WARN SYS FAULT* LED.	Buzzer sounds continuously unless MUTE button is pressed to silence it.	
Alert	Alert instantly INITIATED: ALERT Tones / voice announcements are automatically initiated as per tone generator configuration. If AUTO TEST is active it will turn off immediately ALERT is selected (12A PSU only).	**ALERT**	**Yellow** for *ALERT* LED. Green for *MAINS* LED, Red for *ACTIVE* LED.		

Evacuation	Evacuation instantly INITIATED: EVAC Tones / voice announcements are automatically initiated as per tone generator configuration. If AUTO TEST is active it will turn off immediately EVAC is selected (12A PSU only).	**EVAC**	**Red** for *EVAC* LED. Green for *MAINS* LED, Red for *ACTIVE* LED.		
Public Address	. Operate the *press-to-talk* switch on the microphone to broadcast an announcement, Note: AUTO TEST LED will turn off when PTT is pressed on the microphone (12A PSU only).	**AUTO** 	Green for *MAINS* LED, Yellow for *AUTO TEST,* Red for *ACTIVE* LED.		
Condition:	**Description:**	**Mode Switch:**	**LEDs display:**	**Buzzer:**	**Control Panel LOOKS like this:**
FAULT is detected in • the *BOWS Controller*, or • its tone generator.	The *BOWS Controller* detects a fault in the BOWS system, and it illuminates the *WARN SYS FAULT* LED.	**Any mode setting.**	LEDs illuminated vary per mode active when fault occurs and per type of fault.	Buzzer sounds continuously unless MUTE button is pressed to silence it.	

Cabinet door is closed with mode switch set to ISOLATE, ALERT or EVAC.	The *BOWS Controller* cabinet door may not be closed in any mode other than AUTO. If done, the WARN *SYS FAULT* LED flashes—and a warning buzzer sounds continuously.	**ISOLATE, ALERT or EVAC.**		Buzzer sounds repeatedly in a 4-beep sequence—even if the *MUTE* button has been pressed.	Any one of the illustrations for • **ISOLATE**, • **ALERT**, or • **EVAC**

2.4 OPERATION IN MODE 1—MODE SWITCH ON THE PCB SET TO 1

When Mode "1" is selected the BOWS system will only operate an "EVAC" message.

The Mode of operation is set by a MODE switch (SW3) on the BOWS controller PCB.

The BOWS system can be activated via a Pertronic fire panel RS485 connection to the EVAC Tone generator **OR** a direct connection to the FIP connection on the BOWS controller.

Activation via EVAC Tone Gen RS485

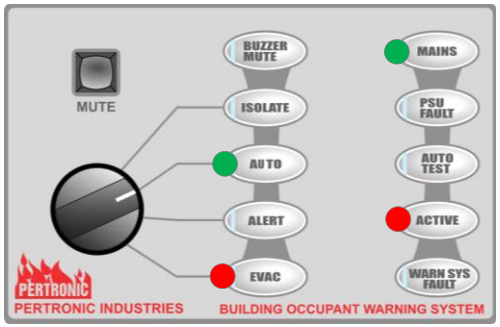
The Pertronic Fire Panel must be programmed to activate the "EVAC" message on the EVAC tone generator.

Activation via EVAC Tone Gen RS485 and BOWS FIP input

The Pertronic Fire Panel must be programmed to activate the "EVAC" message on the EVAC tone generator.

Activation via BOWS FIP input only

No Fire Panel programming or Delay Switch (SW4) setting is required.

Condition:	Description:	Mode Switch:	LEDs display:	Buzzer:	Control Panel LOOKS like this:
Mode 1	The system has been activated by an incoming TGA signal OR activation of the BOWS FIP input and the ALERT or EVAC state is determined by the activating panel and ToneGen configuration. NOTE: The *BOWS* cabinet door is not to be closed in any mode other than AUTO.	**AUTO**	**Green** for *AUTO* LED, **Green** for *MAINS* LED, **Red** for *ACTIVE* and EVAC LED.		

2.5 OPERATION IN MODE 2—MODE SWITCH ON THE PCB SET TO 2

When Mode "2" is selected the BOWS system will:

(a) When activated via EVAC Tone Gen start with an "Alert" message and after a set delay progress to an "Evac" message.
(b) Immediately go to an "EVAC" message whenever a BOWS FIP input is activated

The Mode of operation is set by a MODE switch (SW3) on the BOWS controller PCB.

The BOWS system can be activated via a Pertronic fire panel RS485 connection to the EVAC Tone generator **OR** a direct connection to the FIP connection on the BOWS controller.

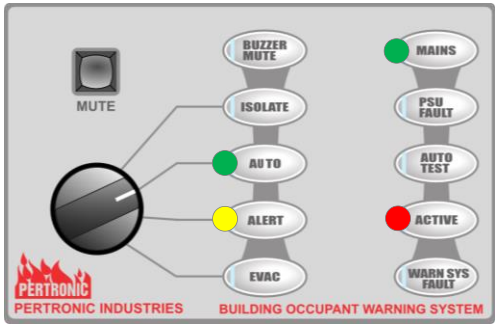
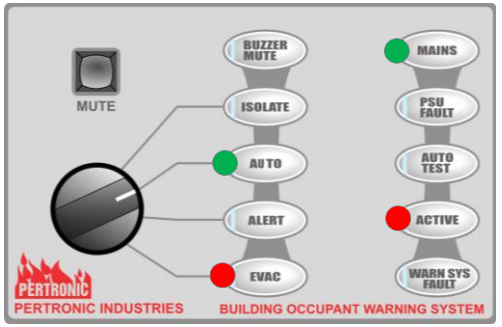
Activation via EVAC Tone Gen RS485 and BOWS FIP input

The Pertronic Fire Panel must be programmed to activate the "Alert" message and after a delay the "Evac" message on the EVAC tone generator.

A delay must also be set using SW4 as detailed above.

Activation via BOWS FIP input only

No Fire Panel programming or Delay Switch (SW4) setting is required.

Condition:	Description:	Mode Switch:	LEDs display:	Buzzer:	Control Panel LOOKS like this:
Mode 2	An incoming TGA signal indicates an ALERT condition. After a set delay the ALERT indication changes to an EVAC indication. Activation of the BOWS FIP input overrides an ALERT indication and puts the BOWS into EVAC NOTE: The *BOWS Controller* cabinet door is not to be closed in any mode other than AUTO.	**AUTO**	**Green** for *AUTO* LED, **Green** for *MAINS* LED, **Red** for *ACTIVE* LED, and **Yellow** for *ALERT* LED <u>until</u> the time set by the DELAY switch expires or FIP input is active — Then, the *ALERT* LED goes off, and the **Red** *EVAC* LED turns on.		FIRST:  and THEN, delay expired or FIP input ACTIVE: 

2.6 INVALID MODE SELECTIONS - (MODE SWITCH ON THE PCB SET TO 3 TO 9)

If an invalid mode is selected during set up the **ALERT** and **EVAC** LED's will flash alternately and the buzzer will sound. This state will remain until a valid mode selection (0-2) is made.

2.7 BOWS FIP INPUT

The BOWS FIP Input is provided for connection to a third party fire panel or equipment. The third party fire panel must be interfaced to the FIP via a relay contact to ensure electrical isolation between the equipment.

Connection to the BOWS FIP input must be terminated with a 10KΩ EOL resistor at the third party equipment end. The input is monitored and if an open-circuit occurs in the wiring to the BOWS FIP input, the BOWS controller will signal a fault condition.

The BOWS FIP input is activated by a short-circuit

2.8 DOOR SWITCH & DOOR INTERLOCK

A BOWS cabinet door switch is usually connected to the BOWS controller to indicate the state of the door (open, closed).

If the user attempts to close the system door when the BOWS is set to **any mode other than AUTO** then a door interlock fault is generated. The yellow **WARN SYS FAULT** LED will flash and the buzzer will sound until the door is re-opened and the mode set to **AUTO**.

A door fault clean contact output is available to connect to external equipment.

2.9 CONNECTING THE WARN SYS FAULT OUTPUT

The BOWS controller board has a common fault relay output for interfacing with a control panel. Voltage free change-over contacts allow connection to any monitored circuit such as an analogue input module or digital input.

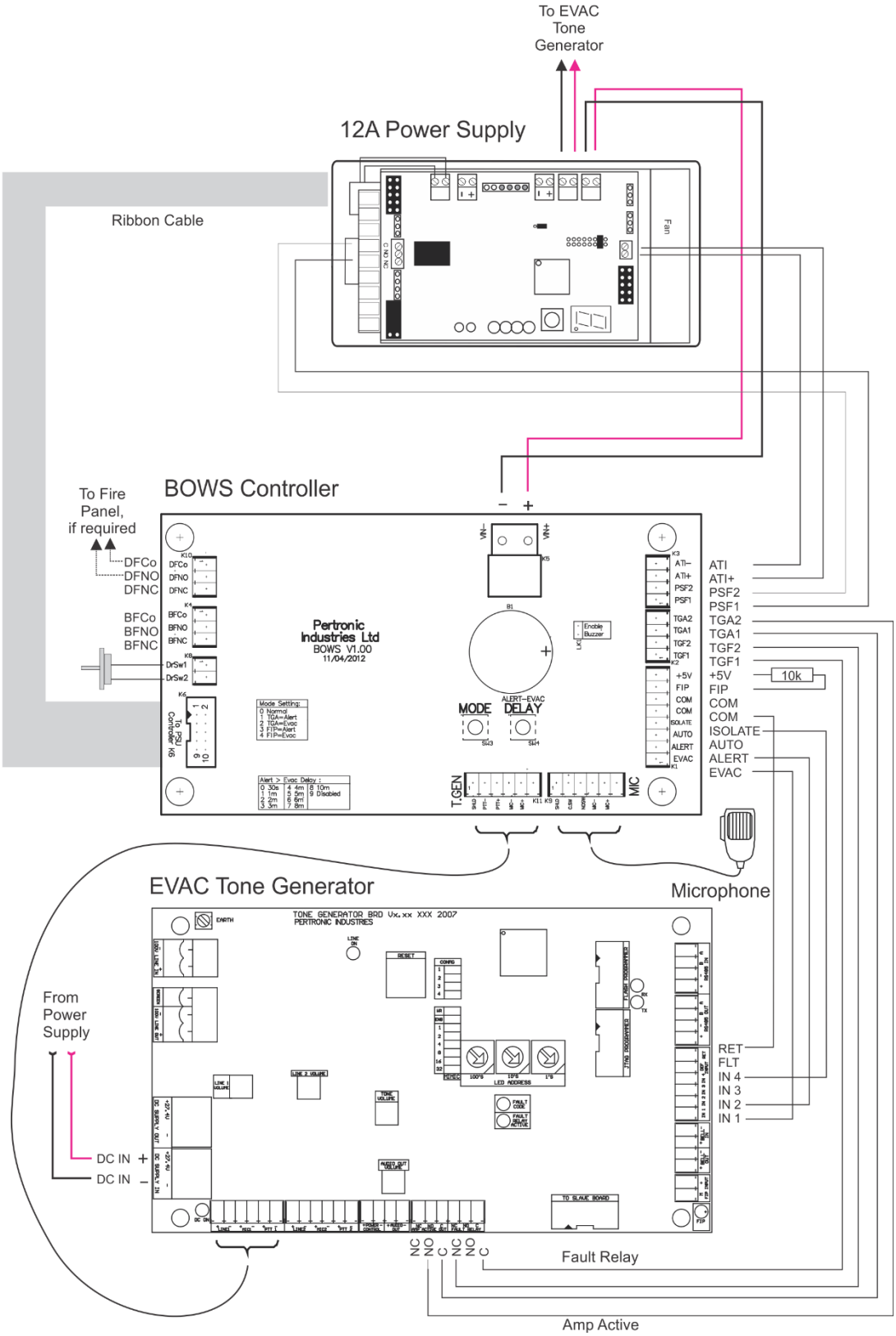
The relay is held energised in its normal condition and is therefore failsafe.

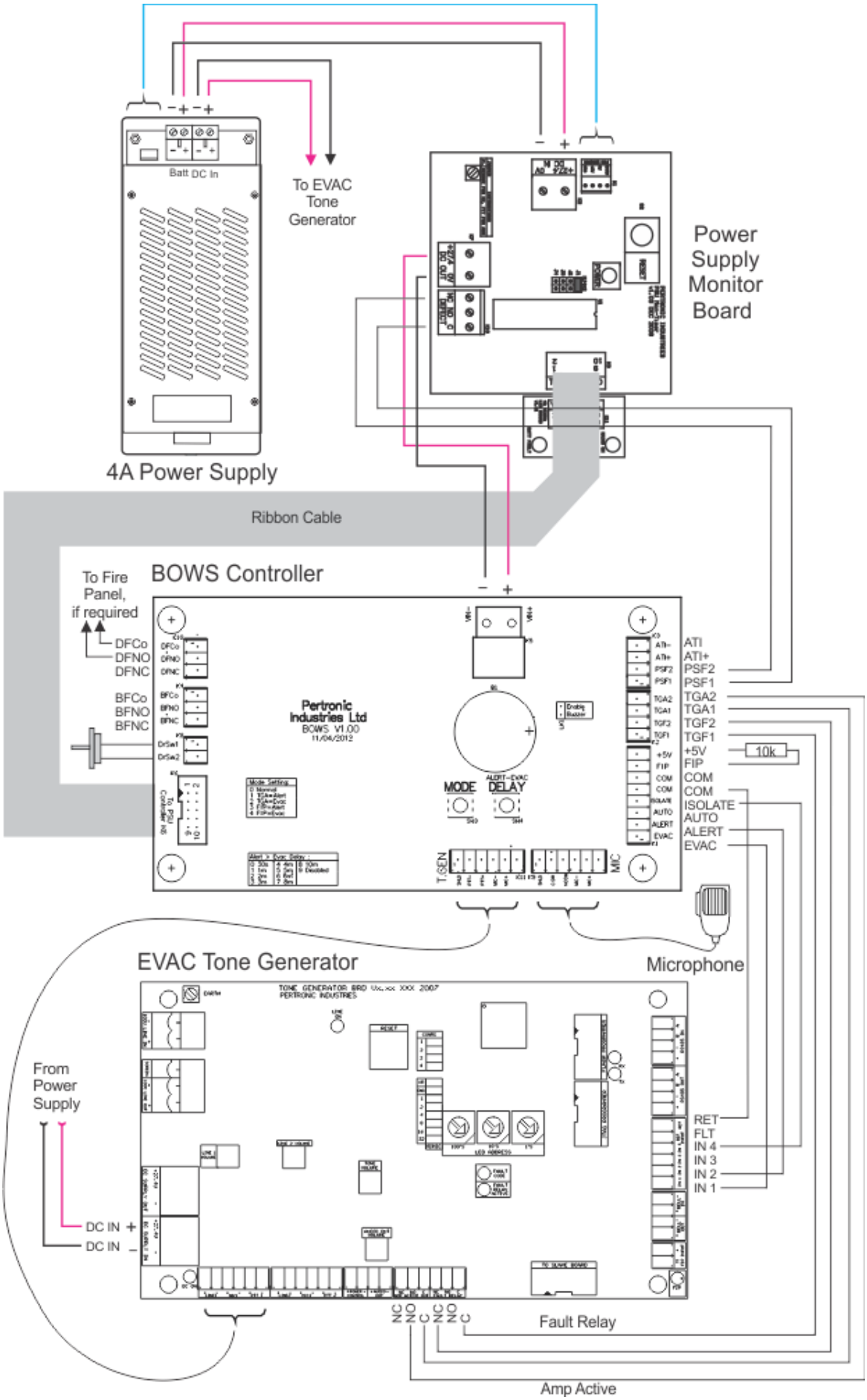
ID	Function
BFCo	Common
BFNC	Closed for fault
BFNO	Open for fault

3. CONNECTIONS

3.1 INTERCONNECTION WIRING DIAGRAM: POWER SUPPLIES AND EVAC

All terminals are clearly marked on the PCB. The power supply LED's input connector is compatible with the Pertronic 12A and 4A power supply controllers/monitors. Both follow:




Batt DC In
To EVAC Tone Generator
Power Supply Monitor Board
4A Power Supply
Ribbon Cable
To Fire Panel, if required
BOWS Controller
DFCo
DFNO
DFNC
BFCo
BFNO
BFNC
Pertronic Industries Ltd
BOWS V1.00
11/04/2012
MODE
DELAY
T.SEN
MIC
ATI
ATI+
PSF2
PSF1
TGA2
TGA1
TGF2
TGF1
+5V
FIP
COM
COM
ISOLATE
AUTO
ALERT
EVAC
10k
EVAC Tone Generator
Microphone
From Power Supply
DC IN +
DC IN –
RET
FLT
IN 4
IN 3
IN 2
IN 1
NC
NO
C
NC
NO
C
Fault Relay
Amp Active

Connector Details

Connector K1	
ID	**Function**
+5	FIP input +ve **Note 1**
FIP	FIP input **Note 1**
COM	Common
COM	Common to Tone Gen
ISOLATE	To Tone Gen Isolate
AUTO	(not used)
ALERT	To Tone Gen Alert
EVAC	To Tone Gen Evac
Note 1. Fit 10KEOL between +5 and FIP	

Connector K2	
ID	**Function**
TGF1	Tone Generator fault output to BOWS
TGF2	Tone Generator fault output to BOWS
TGA1	Tone Generator “Active” output to BOWS
TGA2	Tone Generator “Active” output to BOWS

Connector K3	
ID	**Function**
AT-	Auto-test inhibit –ve (12A PSU only)
AT+	Auto-test inhibit +ve (12A PSU only)
PSF1	From PSU controller relay contact
PSF2	From PSU controller relay contact

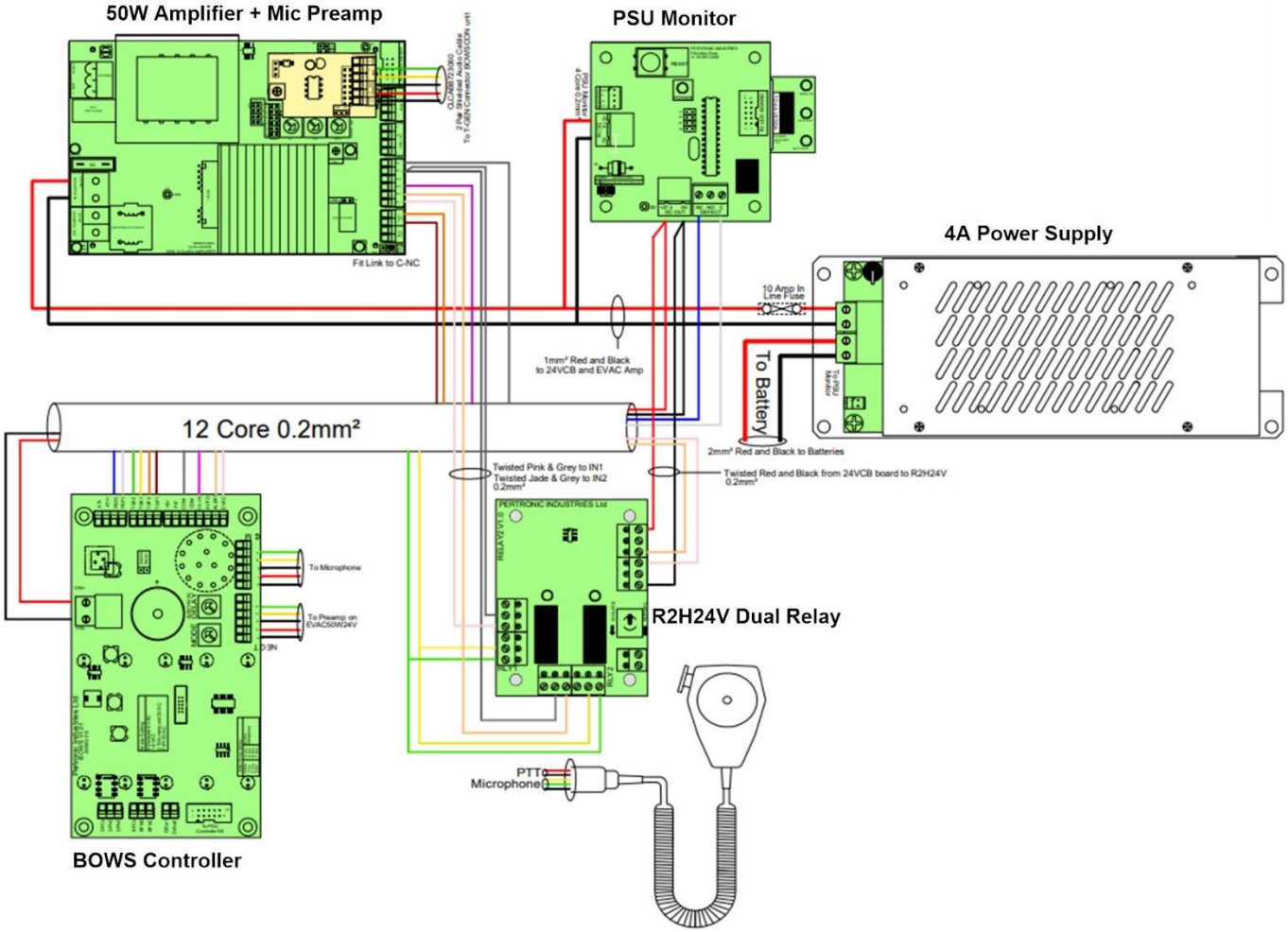
Connector K4	
ID	**Function**
BFCo	BOWS Fault relay common
BFNO	BOWS Fault relay Normally Open
BFNC	BOWS Fault relay Normally Closed
Operated= Normal	**Released=Fault**

Connector K5	
ID	**Function**
VIN+	Supply +ve
VIN-	Supply -ve

Connector K8	
ID	**Function**
DrSw1	Connection to cabinet door switch
DrSw2	Connection to cabinet door switch
Switch Contact closed when door closed	

Connector K10	
ID	**Function**
DFCo	Door Fault relay common
DFNO	Door Fault relay Normally Open
DFNC	Door Fault relay Normally Closed
Relay operates when door closed	

3.2 INTERCONNECTION DIAGRAM: ADDING AN EXTERNAL 50W AMPLIFIER



4. SPECIFICATION

PCB dimensions:	156mm x 86mm.
Mounting holes:	Four-4mm diameter, centred at 40mm x 74mm.
Environmental requirements:	0 to 40°C, 10 to 95% RH (non-condensing).
Indicators:	Mains power, Buzzer muted, Amplifier active, System fault, Power Supply fault, Battery auto-test, Isolated mode, Auto mode, Public Address mode, EVAC mode.
User Controls:	Mode switch (rotary), buzzer mute push-button, and P/A Test push-button.

I/O signals:

TONE GENERATOR CONNECTIONS

Mode outputs:	**Switched to 'COM' when:**
Isolate	'Isolate' selected.
Auto	'Auto' selected.
Alert	'ALERT' selected.
Evacuation	'EVAC' selected.
Mic. Output:	Loop thru from local Microphone.
Inputs:	
Microphone	5-pin with Push-to-talk (PTT) switch.
Tone Gen Fault	Input, Logic 0-5V, 5V=Fault.
Tone Gen Active	Input, Logic 0-5V, 5V=active.

POWER SUPPLY CONNECTIONS

Power requirements	24 VDC, 50mA.
PSU Fault	Input, Logic 0-5V, 5V=fault.
Auto Test Inhibit	Output, Open drain FET, 0V=Inhibit.
(for 12A supply only)	Rating 40V, 500mA.
Mains On LED	Input, Logic 0-5V, Active low.
Battery Fault LED	Input, Logic 0-5V, Active low.
Auto Test LED	Input, Logic 0-5V, Active low.

FIRE PANEL CONNECTION

BOWS fault	Output, Relay:	Closed=normal.
	Contact rating:	120VAC/24VDC. 1.0A max.
Door Interlock fault	Output, Relay:	Closed=normal.
	Contact rating:	120VAC/24VDC. (1.0A max)

FIP INPUT

BOWS FIP input	Activated by clean contact closure Monitored- 10K Ω EOL

DOOR SWITCH INPUT

Open circuit voltage	5V.
Contact format	Contacts closed when door closed.
STANDBY CURRENT	(3A Charger)
50W amplifier	200mA
120W amplifier	350mA

5. DOCUMENT CHANGE HISTORY

Issue Number	Date	Reason for Update	Description of Changes	Author
Issue 1	20/6/2012	Initial Release		JasonM RDB
Issue 1.1	30/10/2012	Standby current info missing	Added to section 4	PF, RDB
Issue 1.2	09/12/2020	CN2909	Added wiring diagram section 3	RJK